Prompt Ladder

Baseline (Weak Prompt)

Prompt

Help me write my ML research question.

Representative Output

Your research question could be about machine learning and SEO. Explain your problem, collect data, train a model, and evaluate it.

Notes

What changed?

None. This is the baseline.

What improved?

Nothing yet.

What still failed?

The answer is generic and doesn't relate to my internship, dataset, or project.

What next?

Give the AI a clearer goal.

Version 1 — Layer: Clear Goal

Prompt

Help me write a research question for my FlyRank ML internship notebook.

Representative Output

The assistant suggested creating a research question about predicting content performance using machine learning.

Notes

Layer Added

Clear goal.

What improved?

The response became relevant to the internship instead of giving general ML advice.

What still failed?

It still didn't know my audience or the business problem.

Next

Specify who will use the work.

Version 2 — Layer: Audience

Prompt

Help me write a research question for my FlyRank ML internship notebook. The audience is an SEO manager who needs to decide which pages to review first.

Representative Output

The assistant focused on helping SEO managers prioritize content for review instead of discussing machine learning in general.

Notes

Layer Added

Audience.

What improved?

The response became more practical because it explained who benefits from the model.

What still failed?

It still lacked details about the available data.

Next

Add real context.

Version 3 — Layer: Context

Prompt

Help me write a research question for my FlyRank ML internship. I am using an anonymized SEO dataset containing impressions, clicks, CTR, average position, content age, and update history. The audience is an SEO manager who needs to decide which pages to refresh first.

Representative Output

The assistant suggested building a decision-support model using historical search signals instead of relying on manual review.

Notes

Layer Added

Context.

What improved?

The answer now referenced the actual dataset and suggested features relevant to the problem.

What still failed?

The structure was difficult to copy directly into the notebook.

Next

Specify the output format.

Version 4 — Layer: Output Format

Prompt

Write the answer using these sections:

- Research Question

- Decision

- Action

- Cost of a Wrong Recommendation

- Why Machine Learning Helps

Representative Output

The response was organized into the requested headings.

Notes

Layer Added

Output format.

What improved?

The answer became much easier to paste directly into the notebook.

What still failed?

Some wording sounded like generic AI writing.

Next

Add quality criteria.

Version 5 — Layer: Quality Criteria

Prompt

Write the notebook using these headings:

- Research Question

- Decision
- Action
- Cost of a Wrong Recommendation
- Why Machine Learning Helps

Requirements:

- Use simple English.
- Avoid buzzwords.
- Do not exaggerate.
- Use careful scientific language.
- Keep it under 250 words.
- Make it sound like a Software Engineering student wrote it.

Representative Output

The response became concise, realistic, and closely matched my own writing style.

Notes

Layer Added

Quality criteria.

What improved?

The writing became more natural and believable while avoiding exaggerated claims.

What still failed?

The first draft was slightly too formal, so I still made a few manual edits before submitting.

This is my honest "didn't fully help" moment. Even with better prompting, I still needed to edit the wording to match my own voice.

Final Reusable Prompt

Help me write a research question for an ML internship notebook.

Context:

I am using an anonymized SEO dataset containing impressions, clicks, CTR, average search position, content age, and update history. My project aims to help SEO managers prioritize which content pages should be reviewed first.

Write the response using these headings:

1. Research Question
2. Decision
3. Action
4. Cost of a Wrong Recommendation
5. Why Machine Learning Helps

Requirements:

- Use simple, professional English.
- Avoid buzzwords and marketing language.
- Make only careful, evidence-based claims.
- Keep the response under 250 words.
- Write as if it were prepared by a Software Engineering student documenting an ML project.